

Stock Market Analysis in SQL

Golden-cross signals on six NSE stocks, and the data trap that changes the answer

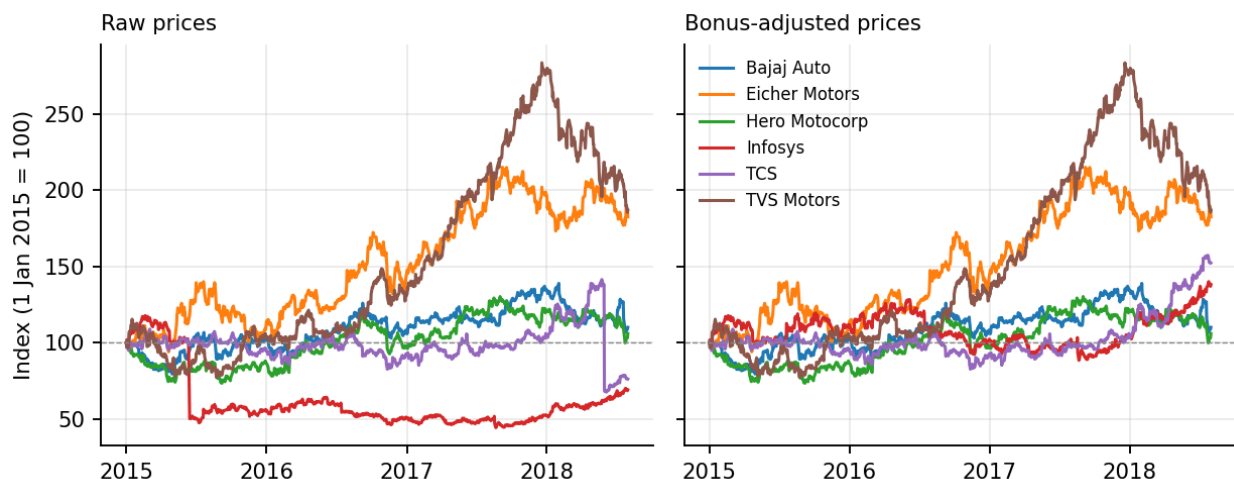
Data: Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS, TVS Motors | 889 trading days each | 1 Jan 2015 to 31 Jul 2018 | Tools: SQL (window functions, CTEs), Python, Streamlit

Executive summary

- **Signals reproduced exactly.** A 20/50-day golden-cross rule gives **56 Buy and 57 Sell** signals across the six stocks. Counts, latest-signal dates and trends all match the course deck.
- **The data hides a trap.** TCS (31 May 2018) and Infosys (15 Jun 2015) each show a one-day drop of about 50%. Both are **1:1 bonus issues**: the share count doubled and the price halved. No value was lost.
- **Adjusting reverses two conclusions.** Raw returns say TCS -23.8% and Infosys -30.9%. Adjusted, they are **+52.4%** and **+38.2%**. No stock lost money over the period.
- **The TCS call flips.** On raw prices TCS's latest signal is a Sell (5 Jun 2018). On adjusted prices it is a **Buy (20 Apr 2018)**. The deck's 'sell TCS' is an artefact of the unadjusted bonus issue.
- **Signals did not beat buy-and-hold.** The strategy trailed buy-and-hold on all six stocks (no costs included). Treat crossovers as one input, not a standalone system.

Stock	Raw return	Adjusted return	Latest signal (adjusted)	Verdict
TVS Motors	+86.9%	+86.9%	Sell, 17 May 2018	Sell / reduce
Eicher Motors	+82.6%	+82.6%	Sell, 06 Jun 2018	Sell / reduce
TCS	-23.8%	+52.4%	Buy, 20 Apr 2018	Buy (deck: Sell)
Infosys	-30.9%	+38.2%	Buy, 07 May 2018	Buy
Bajaj Auto	+10.0%	+10.0%	Buy, 21 Jun 2018	Buy
Hero Motocorp	+6.0%	+6.0%	Sell, 22 May 2018	Sell

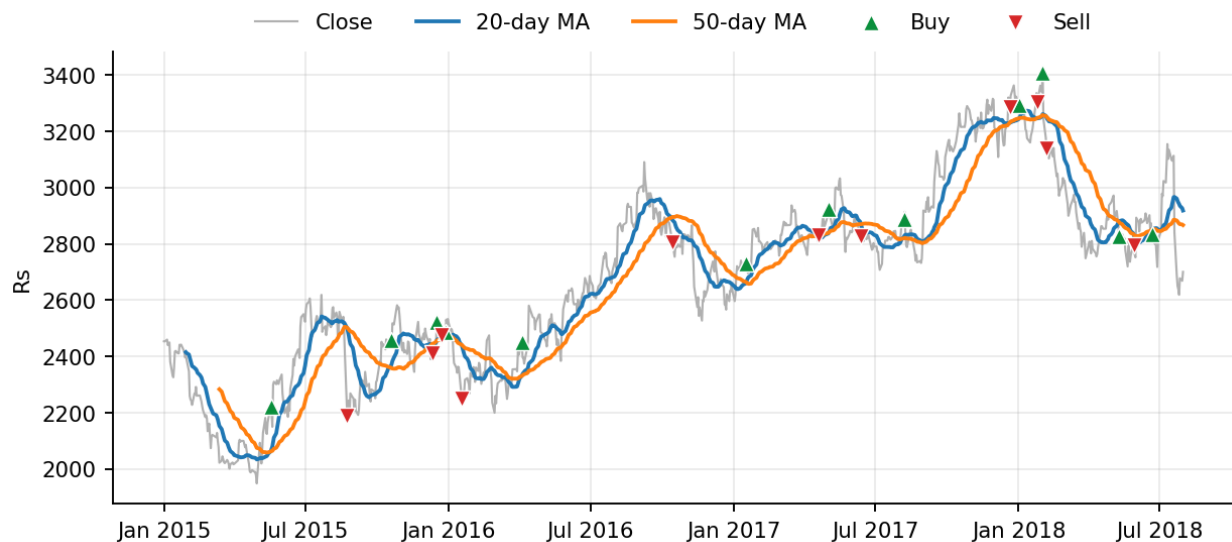
Verdicts follow the latest golden-cross direction on adjusted prices. Sell / reduce for Eicher and TVS is a trend call, not a value call: both have the highest 3.6-year returns.



1. Method: moving averages and golden-cross signals

All analysis is written in SQL. Each stock's close price is smoothed with a **20-day** and a **50-day** moving average using window functions (`AVG() OVER (... ROWS BETWEEN 19 PRECEDING AND CURRENT ROW)`). The averages stay empty until a full window exists, because a partial window is not a real average and would create fake signals.

A **Buy** is issued on the single day the 20-day average moves from at-or-below to above the 50-day average; a **Sell** on the reverse. A crossover is about *two days* (today and yesterday, via `LAG()`), not one; testing only today would mark hundreds of days as Buy.



Bajaj Auto: 12 Buys and 11 Sells. First Buy 18 May 2015, first Sell 24 Aug 2015, latest signal a Buy on 21 Jun 2018 (trend up).

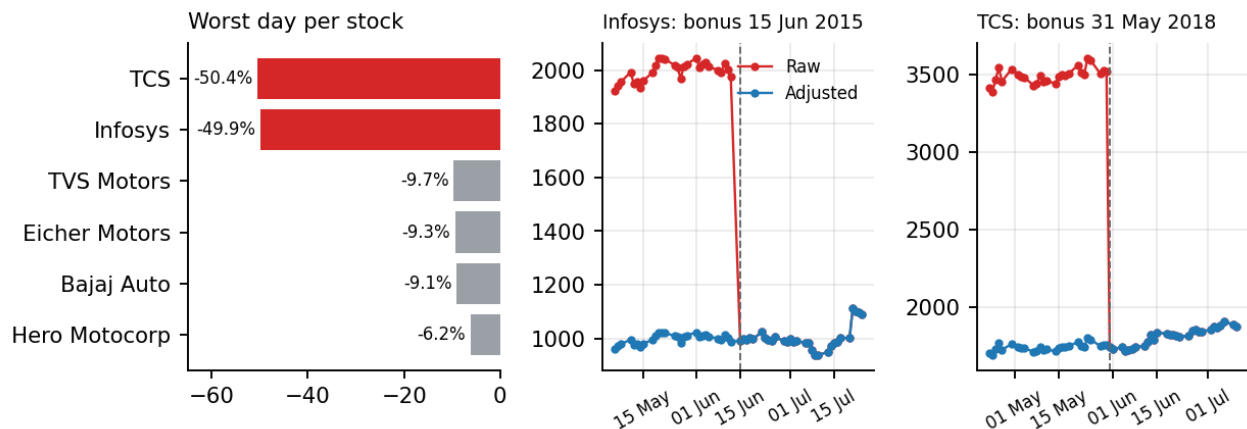
Signals across all six stocks (raw prices, as in the course deck)

Stock	Buys	Sells	Last signal date	Last signal	Trend
Bajaj Auto	12	11	21 Jun 2018	Buy	Shifting up
Eicher Motors	6	7	06 Jun 2018	Sell	Shifting down
Hero Motocorp	9	9	22 May 2018	Sell	Shifting down
Infosys	9	9	07 May 2018	Buy	Shifting up
TCS	12	13	05 Jun 2018	Sell	Shifting down
TVS Motors	8	8	17 May 2018	Sell	Shifting down
Total	56	57			

Data quality: `deliverable_qty` is missing on just two dates (9 Dec 2015 for Eicher, Hero, TCS, TVS; 31 Aug 2017 for Bajaj, Infosys). Four unrelated companies share one date, so the gap is the exchange's reporting, not the companies. It does not touch prices or signals.

2. The data trap: two crashes that never happened

Ranking the raw returns shows TCS at -23.8% and Infosys at -30.9%, two of India's most consistently profitable companies. Looking for each stock's single worst day explains why:

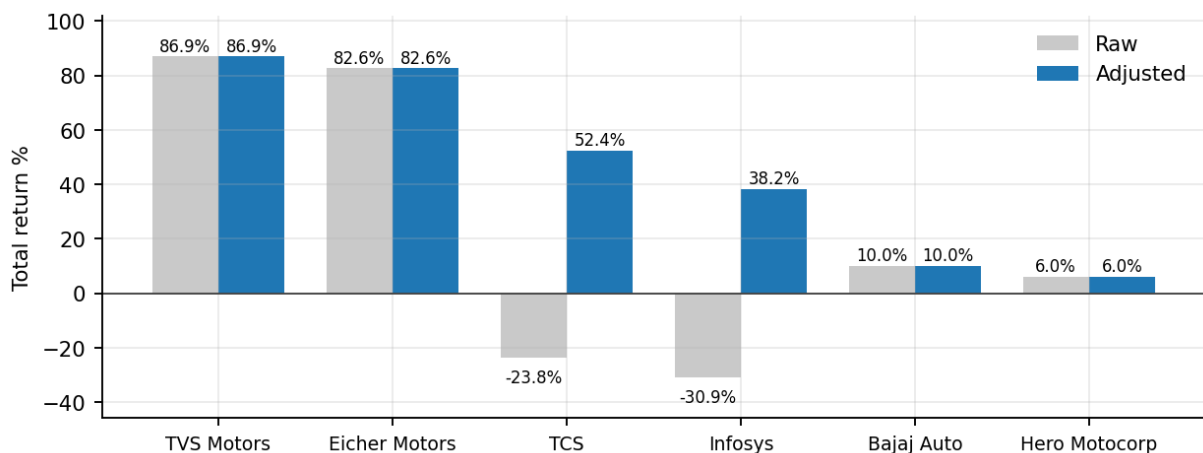


Stock	Worst day	Close (Rs)	Day move	Explanation
TCS	31 May 2018	1,744.80	-50.4%	1:1 bonus issue (price halves)
Infosys	15 Jun 2015	991.10	-49.9%	1:1 bonus issue (price halves)
TVS Motors	03 May 2016	287.90	-9.7%	Market sell-off
Eicher Motors	24 Aug 2015	18,571.75	-9.3%	Market sell-off
Bajaj Auto	24 Aug 2015	2,188.45	-9.1%	Market sell-off
Hero Motocorp	23 Jul 2018	3,163.90	-6.2%	Market sell-off

Four stocks lose 6% to 10% on their worst day. TCS and Infosys lose about half of their value on days with no bad news. In a 1:1 bonus issue every shareholder receives one extra share per share held, so the price halves and nobody gains or loses. Any unadjusted series shows a fake crash, and that corrupts **returns**, **moving averages** and **signals** around the date.

The fix

Divide every price **before** the event date by 2 (Infosys before 15 Jun 2015, TCS before 31 May 2018); prices on and after the event stay unchanged. In SQL this is a CASE inside a CTE.

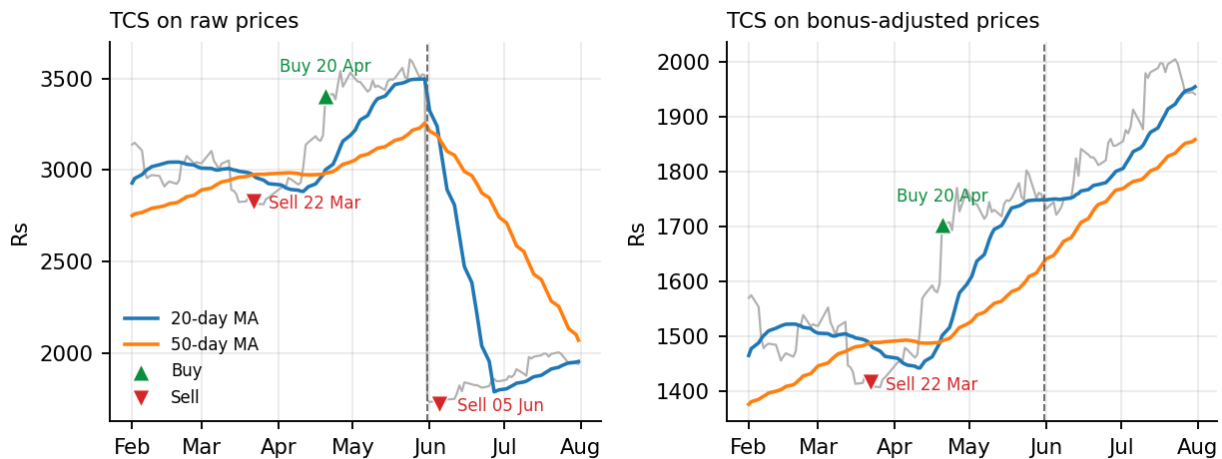


3. What the fix changes

Returns move dramatically: TCS from -23.8% to +52.4% and Infosys from -30.9% to +38.2%. Signals move too, because the averages are built from prices around the bonus date:

Stock	Raw B / S	Adjusted B / S	Latest (raw)	Latest (adjusted)	What happened
TCS	12 / 13	12 / 12	Sell 05 Jun 18	Buy 20 Apr 18	Bonus-day drop drags the 20-day average below the 50-day: a fake Sell. Adjusted: latest signal is a Buy.
Infosys	9 / 9	10 / 10	Buy 07 May 18	Buy 07 May 18	2015 bonus jump hid a real Buy and created a fake Sell. Latest signal unchanged.

The other four stocks have no corporate action in the window, so nothing changes for them.



Dashed line: TCS bonus-issue date (31 May 2018). On raw prices the drop drags the 20-day average under the 50-day and triggers a Sell on 5 Jun; on adjusted prices no such cross occurs and the latest signal remains the Buy of 20 Apr.

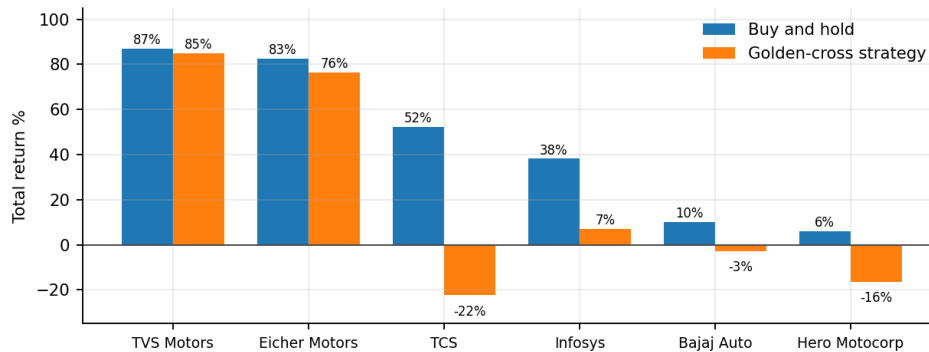
Recommendation

Stock	Adj. return	Latest (adjusted)	Call	Reasoning
Bajaj Auto	+10.0%	Buy, 21 Jun 2018	Buy	Fresh Buy on 21 Jun 2018; modest but positive history.
Infosys	+38.2%	Buy, 07 May 2018	Buy	Buy since 7 May 2018; +38% adjusted, not the -31% the raw data shows.
TCS	+52.4%	Buy, 20 Apr 2018	Buy (deck: Sell)	Adjusted data gives a Buy (20 Apr 2018) and +52%; the deck's Sell is a data artefact.
Eicher Motors	+82.6%	Sell, 06 Jun 2018	Sell / reduce	Trend turned down 6 Jun 2018 after the 2017 peak; deepest drawdown risk with TVS.
TVS Motors	+86.9%	Sell, 17 May 2018	Sell / reduce	Trend down since 17 May 2018; highest volatility (30.8% annualised).
Hero Motocorp	+6.0%	Sell, 22 May 2018	Sell	Trend down since 22 May 2018 and the weakest return (+6.0%).

Bottom line: buy Bajaj Auto, Infosys and (after correcting the data) TCS; sell or reduce Eicher, TVS and Hero. The course deck reached 'sell TCS' only because its final call rests on unadjusted prices.

4. Did the signals make money?

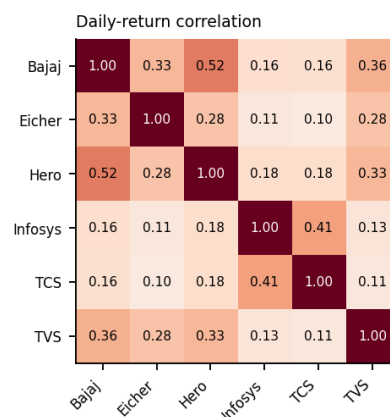
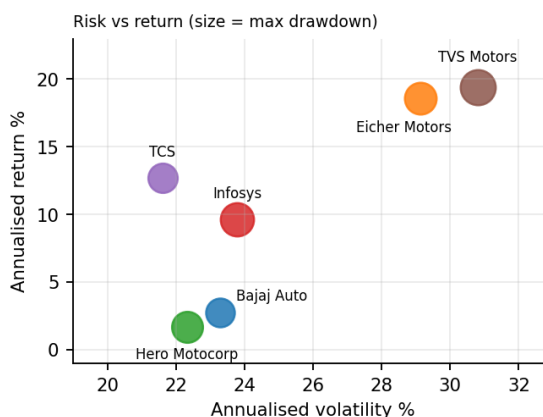
Long-only test on adjusted prices: buy at the close of a Buy day, sell at the close of the next Sell, hold cash in between. Each day's position is decided from the previous close (no look-ahead). Transaction costs and taxes are ignored.



Stock	Buy & hold	Golden cross	Closed trades	Win rate	Avg trade	Time in market
TVS Motors	+86.9%	+85.0%	8	38%	+11.2%	57%
Eicher Motors	+82.6%	+76.5%	6	100%	+10.2%	52%
TCS	+52.4%	-22.4%	11	18%	-3.4%	50%
Infosys	+38.2%	+7.1%	9	33%	-0.4%	54%
Bajaj Auto	+10.0%	-2.7%	11	36%	+0.4%	56%
Hero Motocorp	+6.0%	-16.3%	9	33%	-1.7%	53%

The strategy lagged buy-and-hold on **all six** stocks. Eicher won 6 of 6 trades yet still trailed, because it sat out about half the time during a strong uptrend. TCS is the starkest case: +52.4% buy-and-hold against -22.4% for the strategy, with only 2 of 11 trades profitable. Moving averages lag price, so in rising or choppy markets they buy late and sell late.

5. Risk and diversification



Stock	Ann. return	Ann. volatility	Max drawdown	Return / risk
TVS Motors	19.4%	30.8%	-34.6%	0.63
Eicher Motors	18.6%	29.1%	-28.3%	0.64
TCS	12.7%	21.6%	-24.1%	0.59
Infosys	9.6%	23.8%	-31.1%	0.40
Bajaj Auto	2.8%	23.3%	-23.2%	0.12
Hero Motocorp	1.7%	22.3%	-26.9%	0.07

Eicher and TVS earned the most but suffered the deepest drawdowns (-28% and -35%). TCS delivered the third-highest return with the lowest volatility of the six. Correlations are low (0.10 to 0.52) and lowest between the IT names and the auto names, so mixing the two groups diversifies.

6. Reconciling with the course deck

Signal counts, latest-signal dates and trend directions match the deck for all six stocks. Three of the deck's return lines contain discrepancies:

- **Infosys** is printed as a '3% decrease'. The data gives **-30.9%** (a dropped digit).
- **TCS** lists 2548.20 as the 31 Jul 2018 close and 2454.1 as the 1 Jan 2015 close. In the data 2548.20 is TCS's **1 Jan 2015** close, 1941.25 is the 31 Jul 2018 close, and 2454.1 is **Bajaj's** 1 Jan 2015 close. The -23.8% itself is correct.
- **Eicher's** 82.5% is 82.56% truncated rather than rounded (82.6%).

Limitations

- Moving-average signals are lagging indicators and underperformed buy-and-hold here. They should be combined with other evidence.
- 3.6 years of data in one market regime; results are in-sample, with no costs or taxes. Nothing here is investment advice.
- Only close prices are used. Volume, delivery percentage and volatility filters are natural next steps.
- The two bonus dates were found from price behaviour and confirmed by name. A production pipeline should load a corporate-actions table (splits, bonuses, dividends) instead of hard-coding dates.
- Dividends are not included in returns.

Reproducibility

Everything in this report is generated from the SQL in `sql/analysis_sqlite.sql` (MySQL 8 version: `sql/analysis_mysql.sql`). The notebook walks through each query with interactive charts, and the Streamlit app adds a live SQL playground with 13 practice tasks and an answer checker.